Cardiomyocyte maturation varies across the same cardiac section

a P0 · 21-gene score

n = 17,272

500 µm

b P7 · 21-gene score

n = 30,485

500 µm

c P14 · 21-gene score

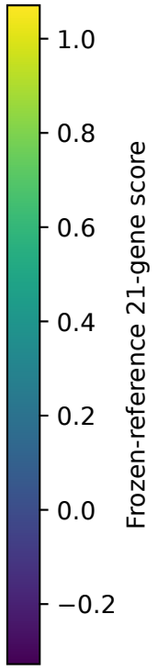
n = 33,659

500 µm

d P21 · 21-gene score

n = 37,949

500 µm



e P0 · rank within CM identity

n = 17,272

500 µm

f P7 · rank within CM identity

n = 30,485

500 µm

g P14 · rank within CM identity

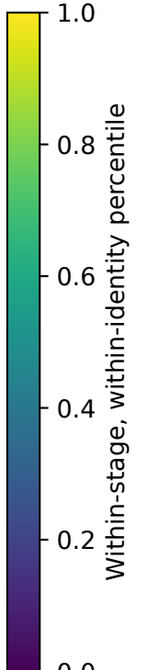
n = 33,659

500 µm

h P21 · rank within CM identity

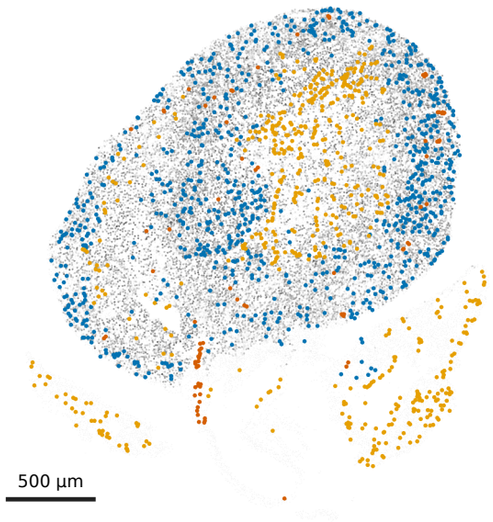
n = 37,949

500 µm

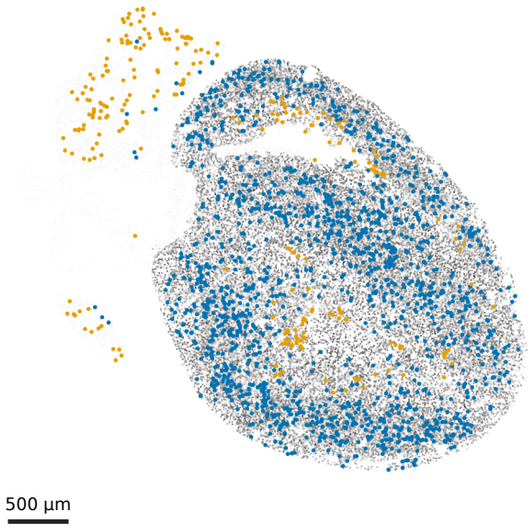


Non-CM identities alongside directly measured CM maturation
Darker cardiomyocytes have higher 21-gene scores; source colors show identity

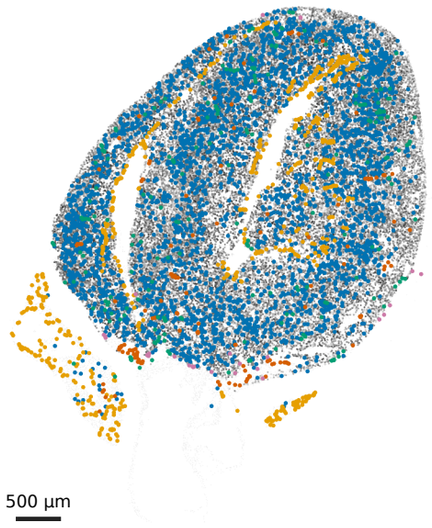
a P0 · endothelial subtypes



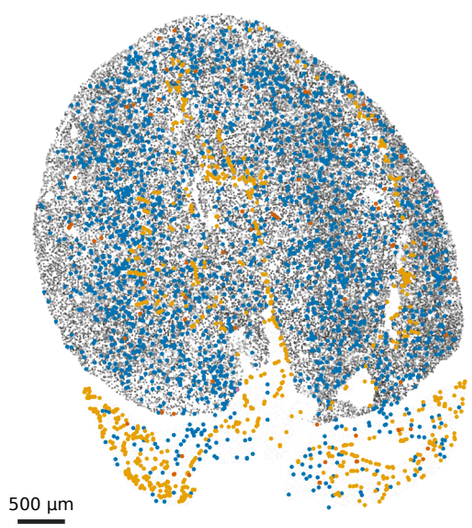
b P7 · endothelial subtypes



c P14 · endothelial subtypes

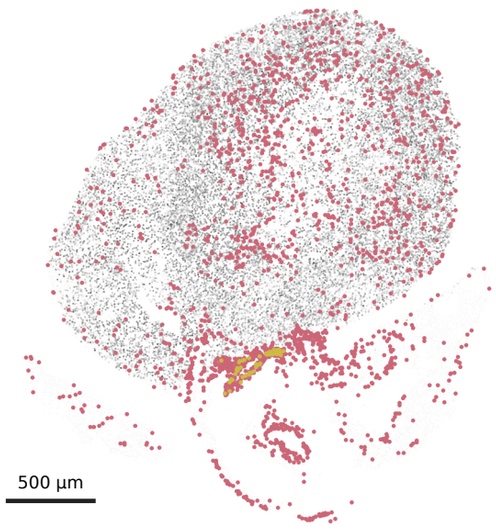


d P21 · endothelial subtypes

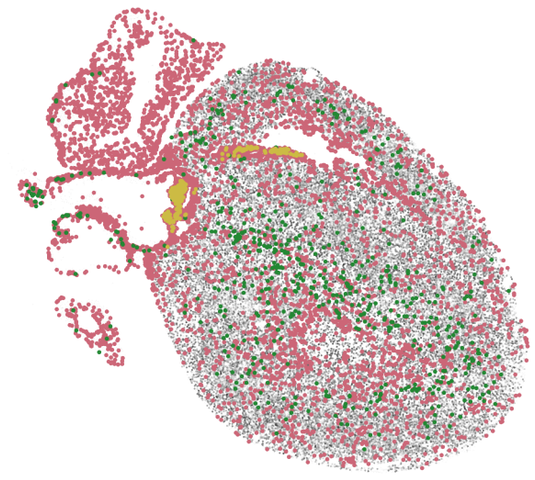


Capillary · Endocardial · Arterial · Venous · Lymphatic

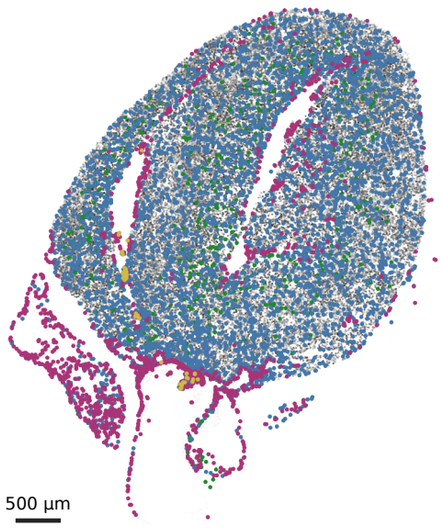
e P0 · fibroblast families



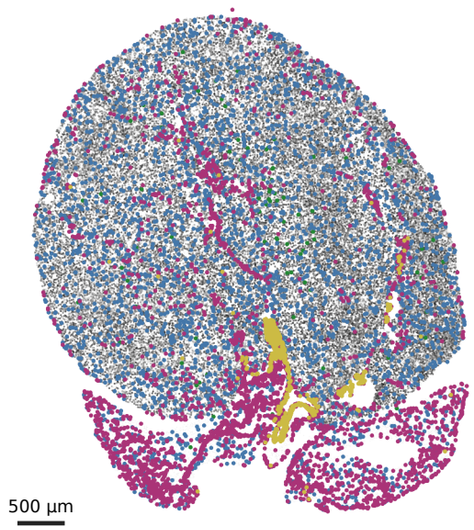
f P7 · fibroblast families



g P14 · fibroblast families

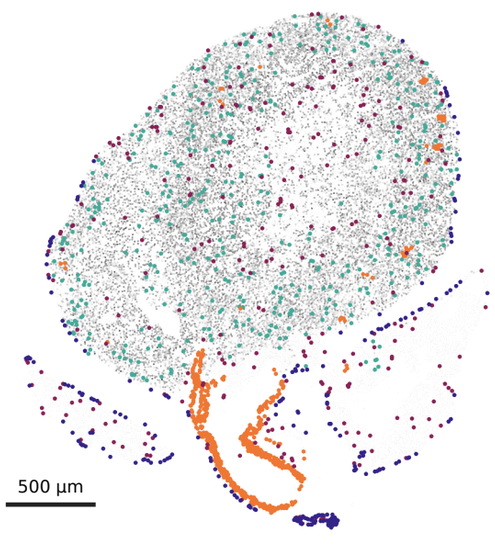


h P21 · fibroblast families

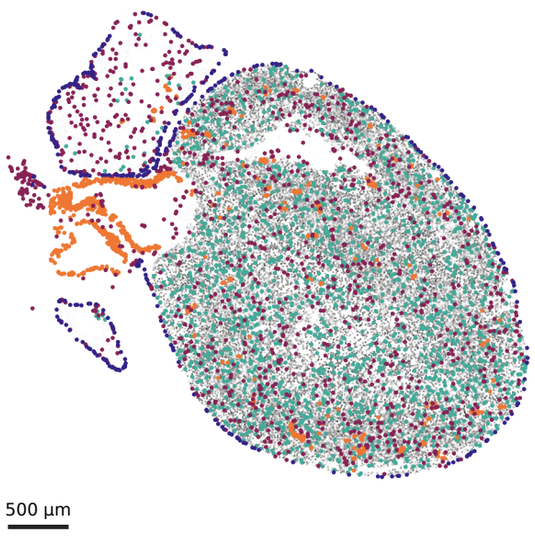


Postn · Gfpt2/Pi16 · Col8a1 · Lsamp · Kirrel3

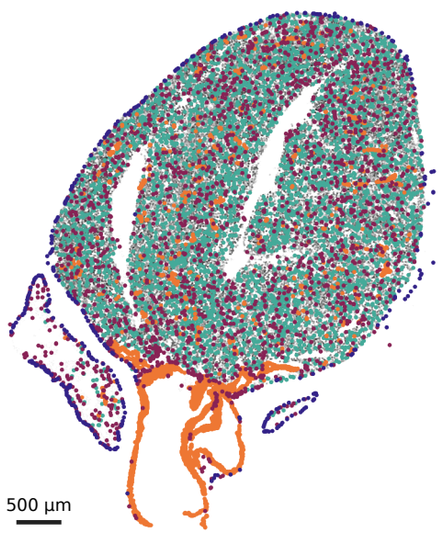
i P0 · other non-CMs



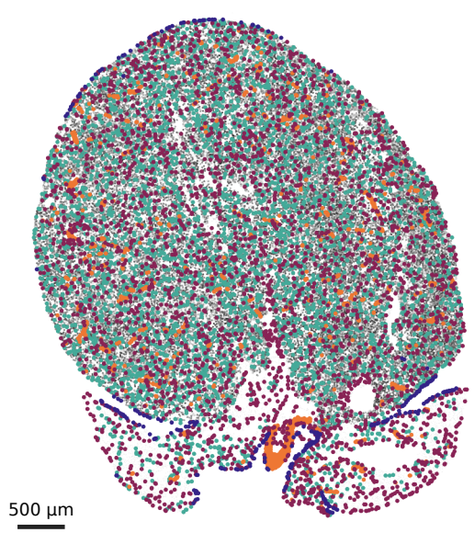
j P7 · other non-CMs



k P14 · other non-CMs



l P21 · other non-CMs



Pericyte · Smooth muscle · Immune · Epicardial

Neighbor relationships differ across CM developmental components
* Global primary BH q < 0.05; gray = unavailable or insufficient support

a Overall 21-gene score

	P0	P7	P14	P21
Endothelial	-0.03	-0.02	-0.01	+0.00
Fibroblast	-0.04	-0.02	-0.02	-0.01
Pericyte	+0.00	-0.00	-0.00	+0.01
Smooth muscle	+0.01	-0.01	-0.00	-0.02*
Immune	-0.01	-0.01	+0.00	+0.01
Epicardial	-0.01	+0.01	+0.00	-0.00
EC: Capillary-like	+0.01	+0.00	-0.00	-0.01
EC: Endocardial-like	-0.01	-0.01	-0.00	-0.01
EC: Arterial-like	-0.00	—	+0.00	+0.00
EC: Venous-like	—	—	-0.00	—
EC: Lymphatic-like	—	—	—	—
FB: Postn	+0.00	-0.01	—	—
FB: Gfpt2/Pi16	—	+0.01	+0.01	+0.00
FB: Col8a1	—	—	-0.00	+0.00
FB: Lsamp	—	—	-0.00	+0.00
FB: Kirrel3	—	+0.00	—	-0.01

b Metabolic-regulatory component

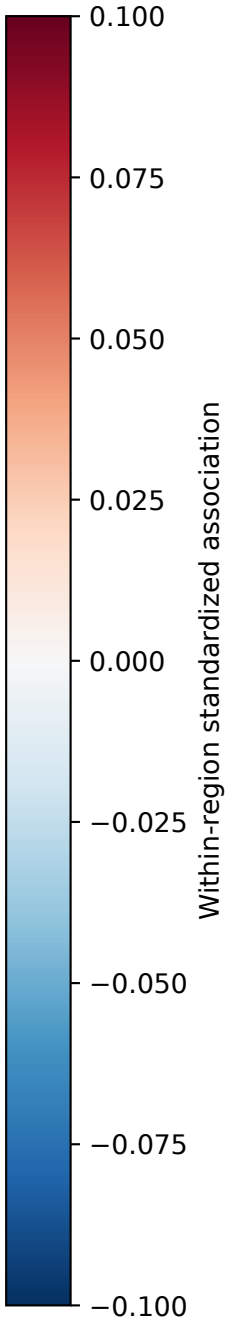
	P0	P7	P14	P21
Endothelial	-0.03	-0.03	-0.02	+0.00
Fibroblast	-0.09*	-0.03	-0.03	-0.02
Pericyte	+0.00	-0.01	-0.01	+0.01
Smooth muscle	+0.01	-0.01	+0.00	-0.01
Immune	-0.01	-0.00	-0.00	+0.00
Epicardial	-0.00	+0.00	+0.00	+0.00
EC: Capillary-like	+0.01	+0.00	+0.00	-0.00
EC: Endocardial-like	-0.01	-0.01	-0.01	-0.01
EC: Arterial-like	-0.01	—	+0.00	-0.00
EC: Venous-like	—	—	-0.00	—
EC: Lymphatic-like	—	—	—	—
FB: Postn	+0.00	-0.01	—	—
FB: Gfpt2/Pi16	—	+0.01	+0.00	+0.00
FB: Col8a1	—	—	+0.00	+0.00
FB: Lsamp	—	—	-0.00	+0.00
FB: Kirrel3	—	+0.01	—	-0.01

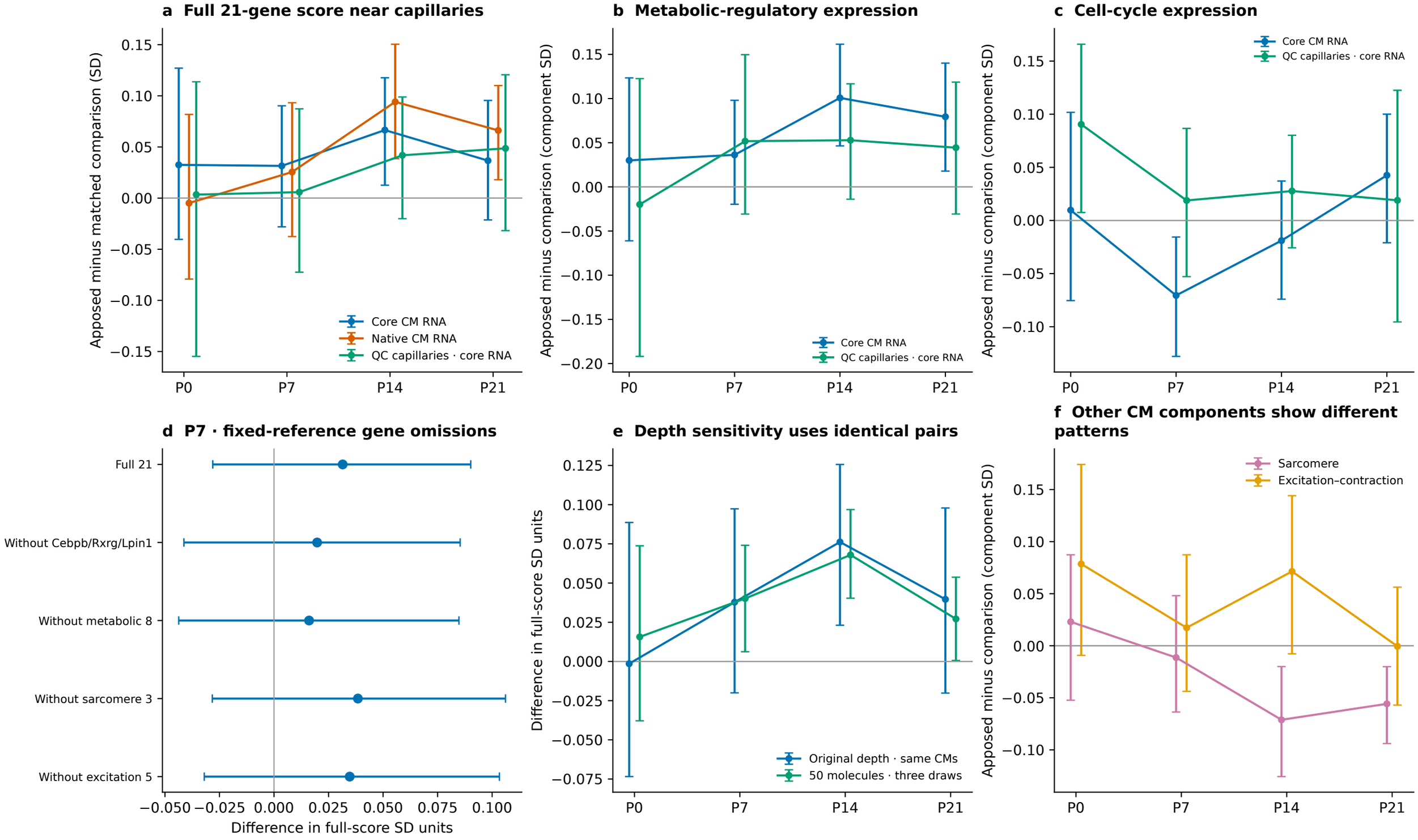
c Sarcomere component

	P0	P7	P14	P21
Endothelial	+0.07	+0.02	-0.03	-0.05
Fibroblast	-0.01	+0.02	-0.06	+0.00
Pericyte	-0.01	+0.01	+0.01	-0.01
Smooth muscle	-0.01	+0.01	-0.01	-0.02
Immune	+0.03	+0.01	-0.00	-0.01
Epicardial	-0.03	+0.01	+0.00	-0.00
EC: Capillary-like	+0.03	-0.01	-0.01	-0.00
EC: Endocardial-like	-0.00	-0.02	+0.02	-0.01
EC: Arterial-like	-0.01	—	+0.01	+0.00
EC: Venous-like	—	—	+0.00	—
EC: Lymphatic-like	—	—	—	—
FB: Postn	+0.00	+0.00	—	—
FB: Gfpt2/Pi16	—	-0.01	+0.00	-0.00
FB: Col8a1	—	—	-0.00	+0.02
FB: Lsamp	—	—	+0.01	-0.03
FB: Kirrel3	—	-0.00	—	+0.01

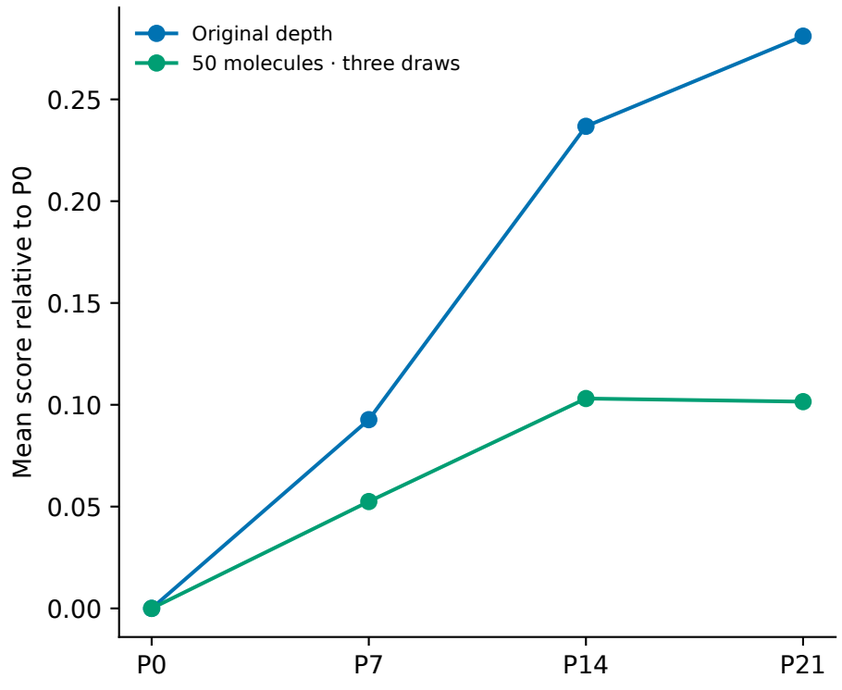
d Cell-cycle expression · separate outcome

	P0	P7	P14	P21
Endothelial	-0.01	-0.00	-0.00	-0.01
Fibroblast	-0.00	-0.00	+0.01	-0.04
Pericyte	+0.00	-0.01	-0.00	-0.01
Smooth muscle	+0.00	+0.00	-0.00	+0.01
Immune	+0.00	-0.01	+0.01	+0.02
Epicardial	+0.01	-0.00	+0.01	-0.01
EC: Capillary-like	+0.01	+0.01	-0.00	+0.01
EC: Endocardial-like	+0.00	-0.00	-0.00	+0.02
EC: Arterial-like	-0.01	—	-0.01	-0.01
EC: Venous-like	—	—	+0.01	—
EC: Lymphatic-like	—	—	—	—
FB: Postn	+0.00	+0.02	—	—
FB: Gfpt2/Pi16	—	-0.02	-0.01	+0.00
FB: Col8a1	—	—	+0.01	-0.00
FB: Lsamp	—	—	+0.01	+0.02
FB: Kirrel3	—	+0.00	—	+0.00

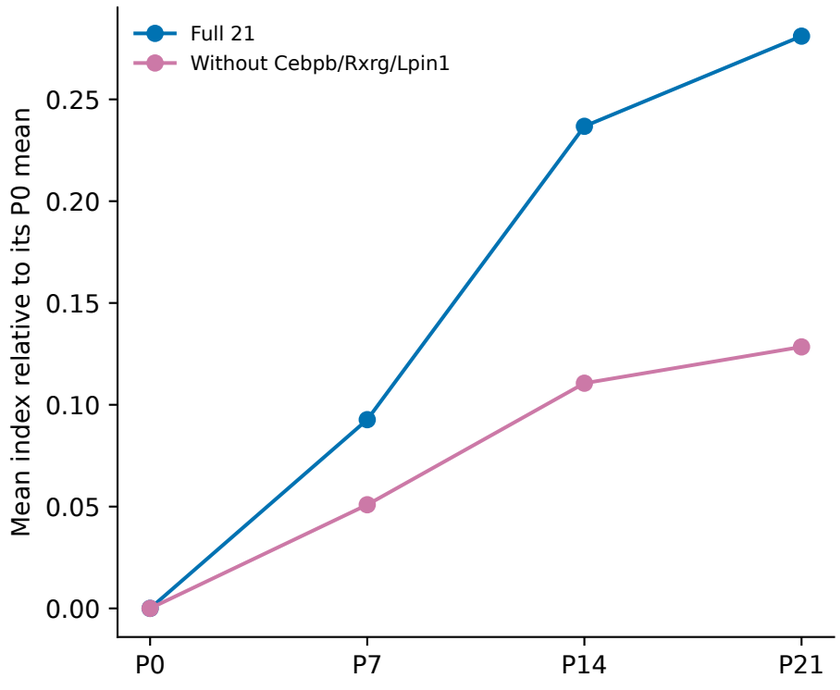



a Full 21-gene score near capillaries
Apposed minus matched comparison (SD)
Core CM RNA
Native CM RNA
QC capillaries · core RNA
P0
P7
P14
P21
b Metabolic-regulatory expression
Apposed minus comparison (component SD)
Core CM RNA
QC capillaries · core RNA
c Cell-cycle expression
d P7 · fixed-reference gene omissions
Full 21
Without Cebpb/Rxrg/Lpin1
Without metabolic 8
Without sarcomere 3
Without excitation 5
Difference in full-score SD units
e Depth sensitivity uses identical pairs
Original depth · same CMs
50 molecules · three draws
f Other CM components show different patterns
Sarcomere
Excitation–contraction

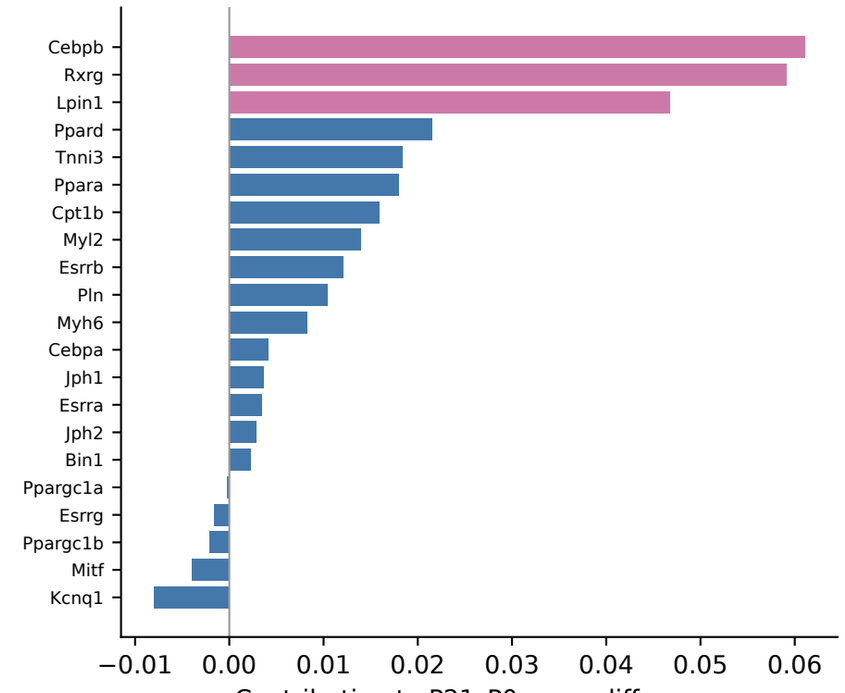
a Age trend persists but weakens at equal depth



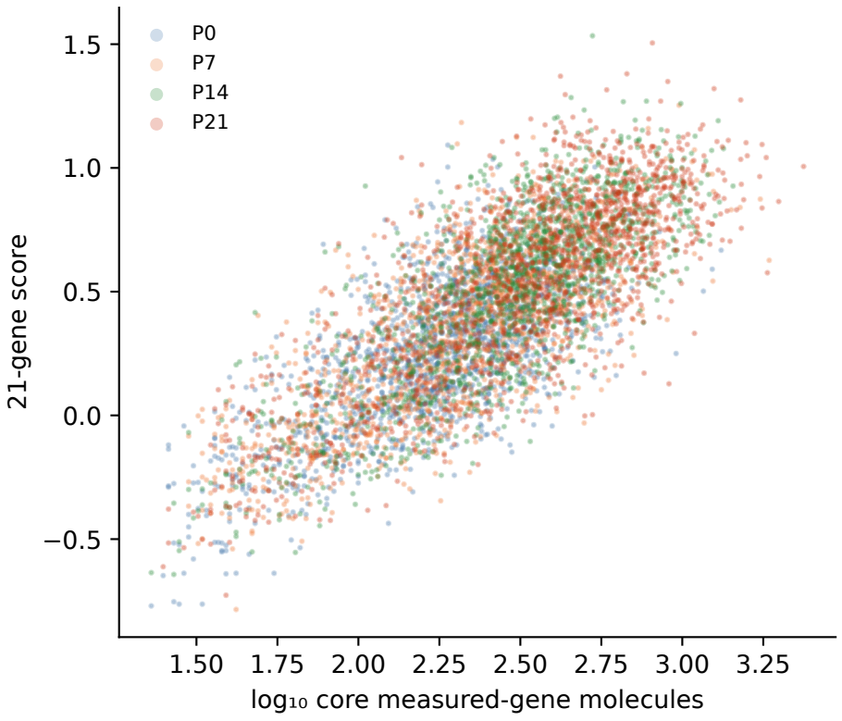
b Joint omission changes the temporal signal



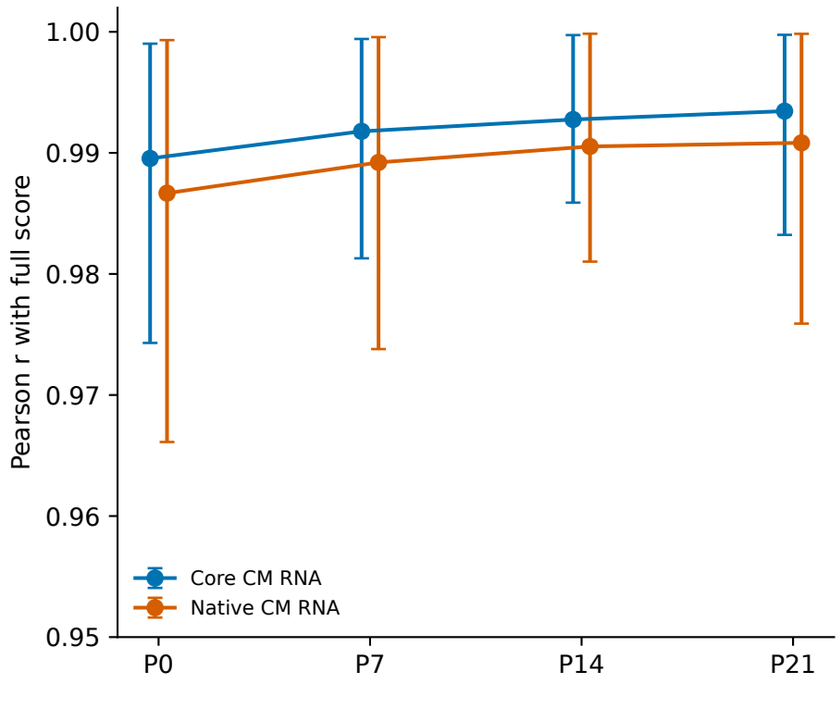
c Temporal contributions are unequal



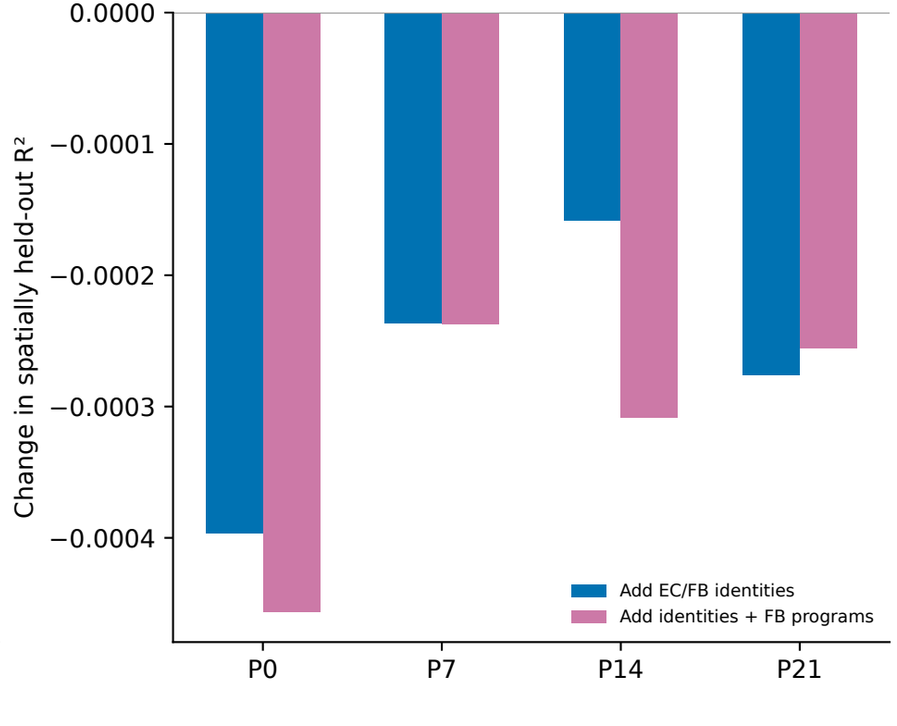
d RNA depth remains an important consideration

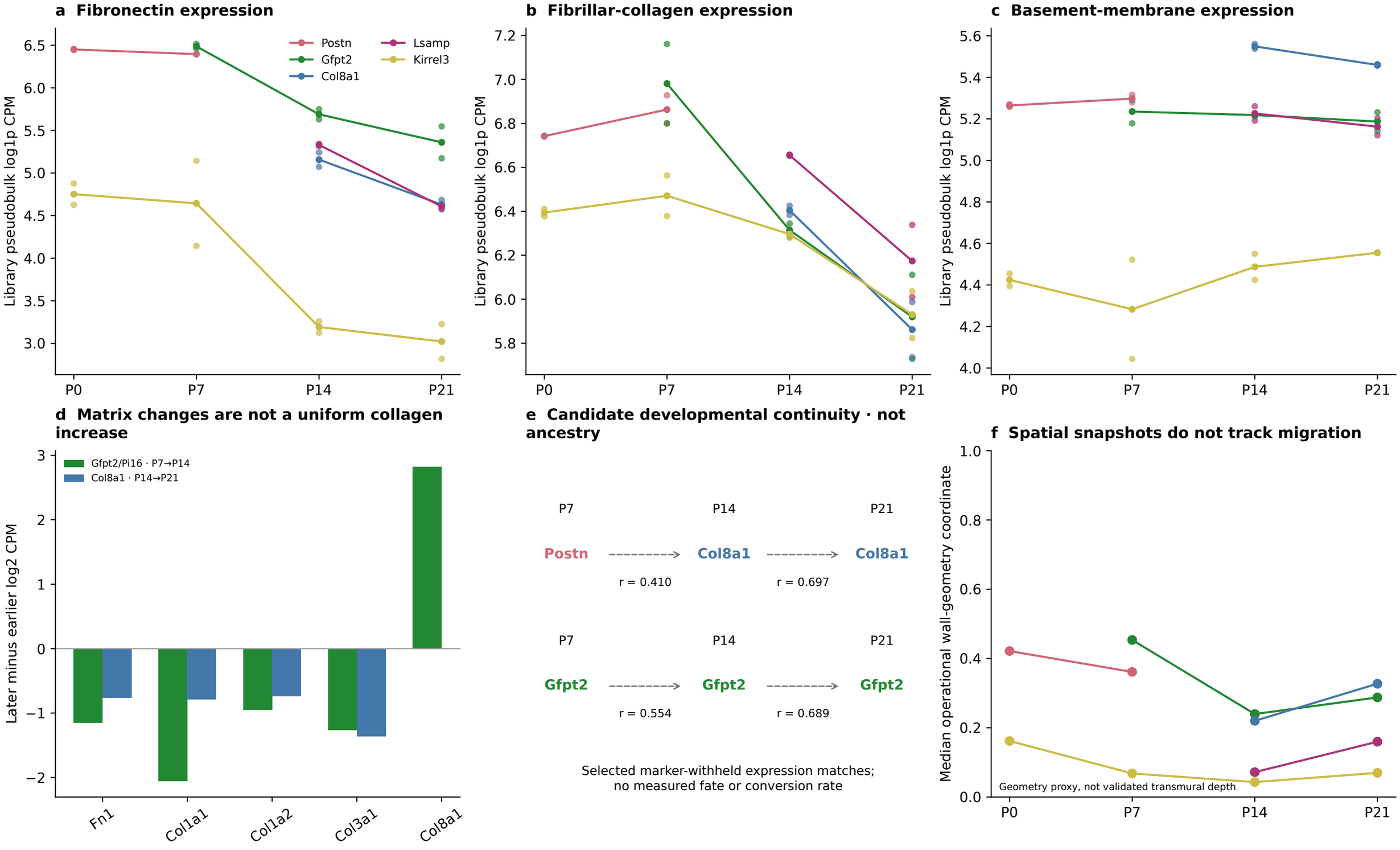


e Single-gene omission preserves score ordering



f Finer labels add no overall-score prediction here




a Fibronectin expression
Postn
Gfpt2
Col8a1
Lsamp
Kirrel3
Library pseudobulk log1p CPM
P0
P7
P14
P21
b Fibrillar-collagen expression
c Basement-membrane expression
d Matrix changes are not a uniform collagen increase
Gfpt2/Pi16 · P7→P14
Col8a1 · P14→P21
Later minus earlier log2 CPM
Fn1
Col1a1
Col1a2
Col3a1
Col8a1
e Candidate developmental continuity · not ancestry
Postn
Col8a1
Col8a1
r = 0.410
r = 0.697
Gfpt2
Gfpt2
Gfpt2
r = 0.554
r = 0.689
Selected marker-withheld expression matches;
no measured fate or conversion rate
f Spatial snapshots do not track migration
Median operational wall-geometry coordinate
Geometry proxy, not validated transmural depth